Assignment 1

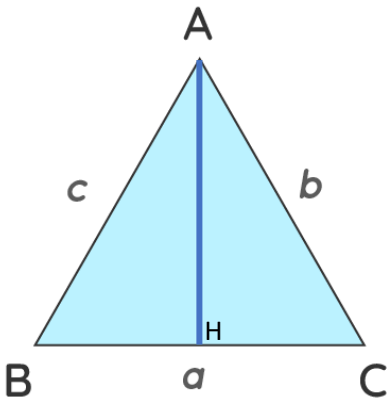
Problem 1

*Use the help command in MATLAB to obtain information about the **dot** and the **norm** commands.*
Let $A(2,5)$, $B(6,1)$, and $C(2,1)$ be three points in the xOy system.

a. Express the vectors $\overrightarrow{CA}$, $\overrightarrow{CB}$, and $\overrightarrow{AB}$ in matrix form.
b. Prove that triangle ABC is a right triangle at C:
   i. Using the inverse Pythagoras theorem (use the command **norm**)
   ii. Using the dot product (use the command **dot**)

Problem 2

Let $A(5.5,5)$, $B(2,3)$, and $C(7,1)$ be three points in the xOy system. Using the **dot** command, find the value of the height $[AH]$.



$$a^2 = b^2 + c^2 - 2bc\cos A$$
$$b^2 = c^2 + a^2 - 2ca\cos B$$
$$c^2 = a^2 + b^2 - 2ab\cos C$$

Problem 3

Create the following matrices:

$$\boldsymbol{A} = \begin{bmatrix} 1 & 8 & 3 \\ 4 & 0 & 6 \\ 7 & 6 & 9 \end{bmatrix};\ \boldsymbol{B} = \begin{bmatrix} 6 & 6 \\ 3 & 9 \end{bmatrix}$$

a. Type a single command which will replace the numbers in columns 1 and 3 of matrix $\boldsymbol{A}$ with the number 1.
b. Type a code which will expand matrix $\boldsymbol{B}$ to a 3×3 matrix, where the added entries contain the value of 2.
c. Combine matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ in a single 3D-array which will be called $\boldsymbol{C}$.

   i. What is the value of $C(1,3,2)$?
   ii. What is the value of $C(1,2,1)$?
d. Concatenate matrix $\boldsymbol{B}$ to the right of matrix $\boldsymbol{A}$. Call the new matrix $\boldsymbol{D}$.
e. Reorder the matrix $\boldsymbol{D}$ as a 9×2 matrix.

Problem 4

Create the 1000×1000 matrix $\boldsymbol{R}$ whose entries are defined as:

$$R_{i,j} = \begin{cases} 1 & for\ (i = j) \cup (i = 1001 - j) \cup (i = 1{,}1000) \cup (j = 1{,}1000) \\ 0 & elsewhere. \end{cases}$$

Hint: Use the command **fliplr**.

Problem 5

Create the following matrix:

$$\boldsymbol{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

a. Create an array $\boldsymbol{B}$ by copying the elements from the second and third columns of array $\boldsymbol{A}$.
b. Create an array $\boldsymbol{C}$ by copying the elements from the first and second rows of array $\boldsymbol{A}$.
c. Use the **max** function to create a vector that contains the maximum values of each **column** of $\boldsymbol{A}$.
d. Use the **min** function to create a vector that contains the minimum values of each **row** of $\boldsymbol{A}$.